

Amazon Sales Analysis

Professional Data Analytics Case Study

Python • MySQL • Power BI • GitHub

Prepared for Data Analyst Portfolio

Executive Summary

Built an end-to-end analytics solution using Python, MySQL, Power BI, and GitHub. The project cleans transactional data, performs SQL-based business analysis, visualizes KPIs in Power BI, and presents business recommendations.

Business Problem

Organizations need a centralized dashboard to monitor sales, customer behavior, product performance, payment methods, and regional sales trends for better decision-making.

Dataset Overview

Dataset: Amazon Sales Dataset

Rows: 10,000

Columns: 21

Main Fields: Order Date, Customer, Product, Category, Brand, Quantity, Unit Price, Discount, Shipping Cost, Total Sales, Payment Method, State, City.

Data Cleaning (Python)

- Removed duplicate records.
- Checked missing values.
- Standardized column names.
- Converted date columns.
- Validated numeric columns.
- Exported cleaned dataset for SQL analysis.

SQL Business Analysis

The SQL phase answered business questions including: Total Sales Total Orders Average Order Value Monthly Sales Trend Sales by Category Sales by Brand Top Products Top Customers Sales by State Payment Method Analysis Order Status Analysis

Power BI Dashboard

The dashboard contains: KPI Cards (Sales, Orders, Quantity, Average Order Value) Monthly Sales Trend Sales by Category Top Products Top Brands Sales by State (Map) Payment Method Distribution Interactive slicers

Key Business Insights

1. Sales performance varies across categories.
2. Product demand changes over time.
3. Certain brands contribute more revenue than others.
4. Geographic analysis highlights high-performing regions.
5. Payment method preferences provide customer behavior insights.

Business Recommendations

- Increase investment in top-performing categories.
- Optimize inventory for high-demand products.
- Expand marketing in high-performing regions.
- Improve customer retention strategies.
- Track KPIs through interactive dashboards.

Skills Demonstrated

Python (Pandas)
MySQL & SQL
Power BI & DAX
Data Cleaning
Data Visualization
Business Analysis
Git & GitHub

GitHub Repository Structure

Amazon-Sales-Analysis/

■■■ data/

■■■ sql/

■■■ dashboard/

■■■ images/

■■■ README.md

■■■ LICENSE

Conclusion

This project demonstrates the complete data analytics lifecycle—from raw data preparation to business intelligence dashboards and actionable recommendations. It is designed to showcase practical skills expected from an entry-level Data Analyst.